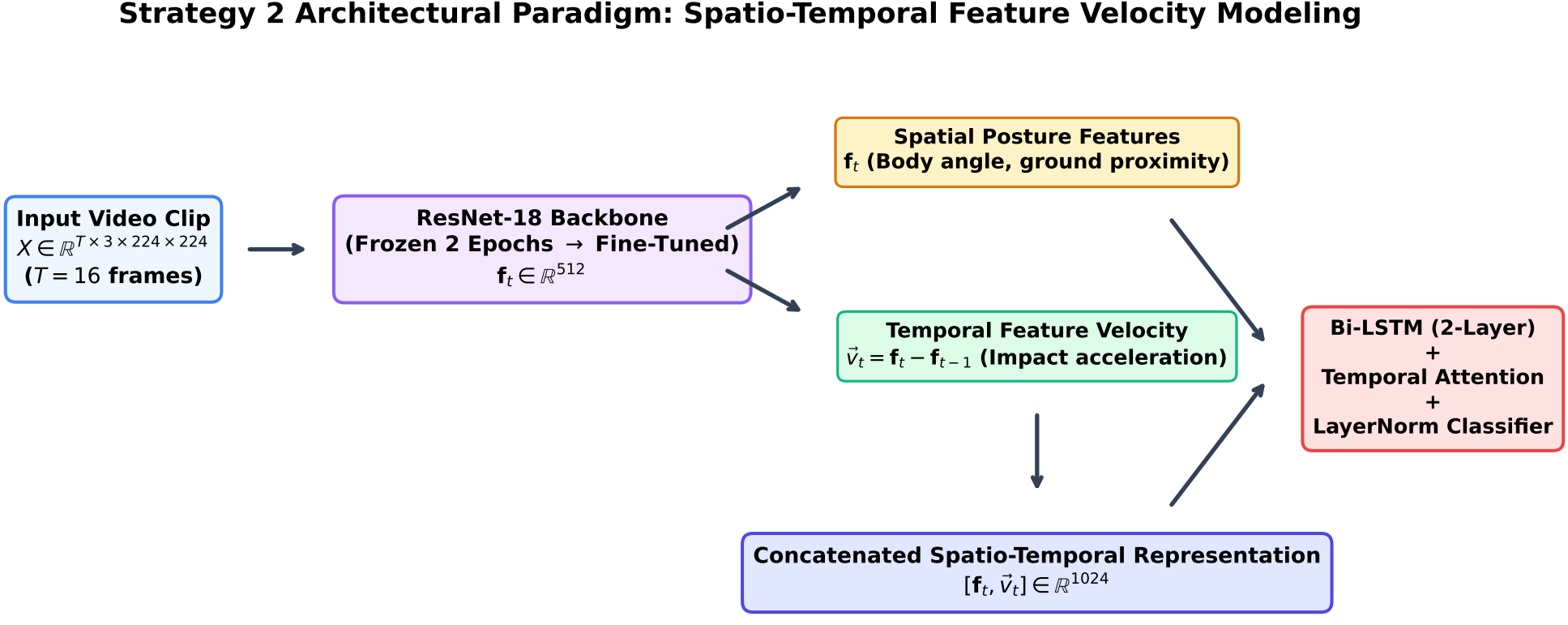
Strategy 2 Architectural Paradigm: Spatio-Temporal Feature Velocity Modeling
Input Video Clip
$X \in \mathbb{R}^{T \times 3 \times 224 \times 224}$
($T = 16$ frames)
ResNet-18 Backbone
(Frozen 2 Epochs → Fine-Tuned)
$\mathbf{f}_t \in \mathbb{R}^{512}$
Spatial Posture Features
$\mathbf{f}_t$ (Body angle, ground proximity)
Temporal Feature Velocity
$\vec{v}_t = \mathbf{f}_t - \mathbf{f}_{t-1}$ (Impact acceleration)
Concatenated Spatio-Temporal Representation
$[\mathbf{f}_t, \vec{v}_t] \in \mathbb{R}^{1024}$
Bi-LSTM (2-Layer)
+
Temporal Attention
+
LayerNorm Classifier